Positive feedback changes time constant of leaky integrator

A single neuron (or single population) firing rate model may take the form,

$$\tau \dot{v} = -v + wv + f \,,$$

where v is firing rate (output), τ is the time constant for changes in firing rate, and f is the drive (input). The firing rate v is a "leaky integrator" of the drive f (small τ is more leaky).

Writing $\bar{v}$ for the steady state firing rate,

$$0 = -\bar{v} + w\bar{v} + f \Rightarrow \bar{v} = \frac{f}{1-w} \,.$$

We see that the synaptic strength w influence the slope of $v(f)$.

The important point for Week 2 is this: the synaptic strength w also influences the time constant of the firing rate v. Furthermore, a long time constants corresponds to an integrator that is less leaky. To see this, defined $\delta v = v - \bar{v}$. The new variable δv is the deviation of the firing rate from steady state. Because $\dot{\delta v} = \dot{v}$, we have

$$\tau \dot{\delta v} = -(\bar{v} + \delta v) + w(\bar{v} + \delta v) + f = -(1-w)(\bar{v} + \delta v) + f$$

$$\tau \dot{\delta v} = -(1-w)\delta v \Rightarrow \underbrace{\frac{\tau}{1-w}}_{\tau_*} \dot{\delta v} = -\delta v$$

The new time constant τ_* is very large as w increases from 0 to 1.